

5.1 dplyr: rowwise() and across()

rowwise() summarizes across columns in one row. across() repeats a function on many columns.

1. rowwise() is a group of one

group_by() groups many rows. rowwise() makes each row its own group so a summary runs **across columns in that row**. ungroup() removes it.

```
df <- tibble(name = c("Mara", "Hadley"), x = 1:2, y = 3:4, z = 5:6)

df |> mutate(m = mean(c(x, y, z)))
# one mean for the whole table

df |>
  rowwise(name) |>
  summarize(m = mean(c(x, y, z)), .groups = "drop")
# one mean per row
```

c_across() selects several columns with the same helpers as select() :

```
df <- tibble(id = 1:6, w = 10:15, x = 20:25, y = 30:35, z = 40:45)
rf <- df |> rowwise(id)

rf |> mutate(total = sum(c_across(w:z)))
rf |> mutate(total = sum(c_across(where(is.numeric))))
```

For speed on a whole frame, rowSums() and rowMeans() avoid the split-and-join:

```
df |> mutate(total = rowSums(across(where(is.numeric))))
```

starwars (in dplyr) is the usual practice table. Without rowwise(), max(height, mass, birth_year) is one number for the whole column stack, repeated on every row.

2. across() repeats a function on columns

across() lives **inside** mutate(), summarize(), filter(), and the other data-masking verbs. It does not change the data by itself.

Two arguments:

- .cols: tidyselect, default everything()

- `.fns`: a function, a named list of functions, or a formula `~ median(.x, na.rm = TRUE)`. `.x` is the current column.

```
starwars |>
  summarize(across(where(is.numeric), ~ median(.x, na.rm = TRUE)), .groups = "drop")

starwars |>
  group_by(species) |>
  filter(n() > 1) |>
  summarize(
    across(c(sex, gender, homeworld), ~ length(unique(.x))),
    n = n()
  )
```

Grouping variables are skipped so you do not summarize them by accident.

A named list makes several summaries per column:

```
min_max <- list(
  min = ~ min(.x, na.rm = TRUE),
  max = ~ max(.x, na.rm = TRUE)
)
starwars |> summarize(across(where(is.numeric), min_max))
```

Order matters. If you compute `n = n()` first, `n` is numeric, so a later `across(where(is.numeric), sd)` will try to take the SD of that constant and give `NA`. Put `n = n()` last.

`across()` replaces the old `*_if()`, `*_at()`, and `*_all` helpers. `select()` and `rename()` already use `tidyselect`; use `rename_with()` to change names with a function.

3. Practice

1. For each `starwars` character, the maximum of `height`, `mass`, and `birth_year` (`na.rm = TRUE`). Who is largest? Try it with and without `rowwise()`.
2. Median of every numeric column, by `species` and `gender`, plus `n()`, sorted by `n` descending.